

**Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus**

Lab Rubrics Form	
Lab Name	Arithmetic Operations and Data Conversion in C Programming
Lab Number	02
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
Modern Tools Usage				
PLO-10	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			
Communication Skills				

Lab 02: Arithmetic Operations and Data Conversion in C Programming

Objective

To understand the use of arithmetic operators, operator precedence, user input, mathematical calculations, unit conversions, and basic number manipulation using C programming.

Learning Outcomes

After completing this lab, the student will be able to:

- Use arithmetic operators in C.
- Take input from the keyboard using scanf().
- Apply operator precedence rules.
- Perform mathematical and unit conversion calculations.
- Manipulate numbers using arithmetic and modulus operators.
- Develop programs for solving simple engineering problems.

In lab Tasks:

1. Perform arithmetic operations on two predefined numbers.
2. Input two numbers from the user and perform arithmetic operations.

Task 1:

Write a program that print two numbers and perform all arithmetic operations.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int P=38;
    int Q=4;
    int S,T,R,N;
    S= P+Q;
    T= P-Q;
    R= P*Q;
    N= P/Q;
    printf("\t%d+%d=%d\n",P,Q,S);
    printf("\t%d-%d=%d\n",P,Q,T);
```

```

printf("\t%d*%d=%d\n",P,Q,R);
printf("\t%d/%d=%d\n",P,Q,N);
return 0;
}

```

Output:

```

"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
38+4=42
38-4=34
38*4=152
38/4=9
Process returned 0 (0x0)   execution time : 0.156 s
Press any key to continue.

```

Task 2:

Write a program that input two numbers and perform all arithmetic operations.

Code:

```

#include <stdio.h>
#include <stdlib.h>

int main()
{
    int P;
    int Q;
    printf("Enter the value of P:\n");
    printf("Enter the value of Q:\n");
    scanf("%d%d",&P, &Q);
    printf("\t%d+%d=%d\n",P,Q,P+Q);
    printf("\t%d-%d=%d\n",P,Q,P-Q);

    printf("\t%d*%d=%d\n",P,Q,P*Q);
    printf("\t%d/%d=%d\n",P,Q,P/Q);
    return 0;
}

```

Output:

```
"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
Enter the value of P:
Enter the value of Q:
42
6
    42+6=48
    42-6=36
    42*6=252
    42/6=7

Process returned 0 (0x0)   execution time : 5.426 s
Press any key to continue.
```

Post Lab Tasks:

1. Evaluate an arithmetic expression using operator precedence.
2. Convert temperature from Centigrade to Fahrenheit.
3. Convert distance from kilometers into meters, centimeters, inches, and feet.
4. Reverse a five-digit number using the modulus operator.
5. Calculate the cost price of one item from the total selling price and profit.

Task 1:

Determine the hierarchy of operations and evaluate the following expressions, assuming that kk is a float variable: $kk = 3/2 * 4 + 3/8$

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    /* Evaluate the following Expression*/
    float kk;
    kk=3/2*4+3/8;
    printf("\nThe value of kk Expression = %f\n",kk);

    return 0;
}
```

Output:

"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"

```
The value of kk Expression = 4.000000
Process returned 0 (0x0)   execution time : 0.109 s
Press any key to continue.
```

Task 2:

Temperature of a city in centigrade degrees is input through the keyboard. Write a program to convert this temperature into Fahrenheit degrees. ($T(^{\circ}\text{F}) = T(^{\circ}\text{C}) \times 1.8 + 32$)....

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int TempF;
    int TempC;
    printf("\n Enter the Temperature in Centigrade\n",TempC);
    scanf("%d", &TempC);
    /* Find Temperature of City in Fahrenhiet */
    TempF=(TempC*1.8+32);
    printf(" Temperature of City in Centigrade=%d\n",TempF);
    return 0;
}
```

Output:

"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"

```
Enter the Temperature in Centigrade
56
Temperature of City in Centigrade=132
Process returned 0 (0x0)   execution time : 4.395 s
Press any key to continue.
_
```

Task 3:

The distance between 2 cities (in km) is input through the keyboard. write a program to convert and print this distance in meters, feet, inches and centimeters.

(Formulas:

```
meter=km*1000;
centimeter=meter*100;
inch=centimeter/2.54;
feet=inch/12;)
```

Code:

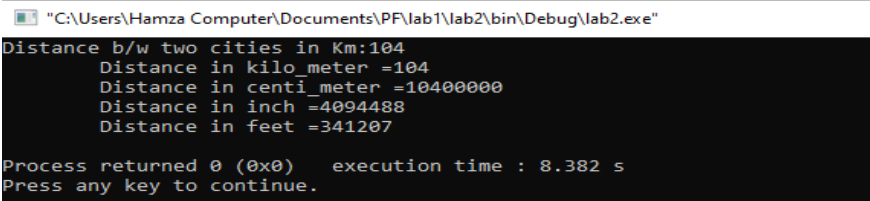
```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int meter,kilo_meter, centi_meter;
    int inch,feet,distance;
    printf("Distance b/w two cities in Km:");
    scanf("%d", &kilo_meter);

    meter= kilo_meter*1000;
    printf("\tDistance in kilo_meter =%d\n", kilo_meter);
    centi_meter= meter*100;
    printf("\tDistance in centi_meter =%d\n", centi_meter);
    inch= centi_meter/2.54;
    printf("\tDistance in inch =%d\n",inch);

    feet= inch/12;
    printf("\tDistance in feet =%d\n", feet);
    return 0;
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
Distance b/w two cities in Km:104
    Distance in kilo_meter =104
    Distance in centi_meter =10400000
    Distance in inch =40944.88
    Distance in feet =3412.07

Process returned 0 (0x0)   execution time : 8.382 s
Press any key to continue.
```

Task 4:

If a 5-digit number is input through the keyboard, write a program to reverse the number.
(hint: use modulus % operator)

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int num, num_1, num_2, num_3, num_4, num_5;
    int num_reverse;
    printf("\n Please enter a five digit number :");
    scanf("%d",&num);
    num_1 = num % 10;
    num /= 10;
    num_2 = num % 10;
    num /=10;
    num_3 = num % 10;
    num = num /10;
    num_4 = num % 10;
    num = num / 10;
    num_5 = num % 10;
    num /= 10;
    num_reverse = num_1 * 10000 + num_2 * 1000 + num_3 * 100 + num_4 * 10 +num_5;
    printf("\n So the Reversed number is : %d\n", num_reverse);

    return 0;
}
```

Output:

Task 5:

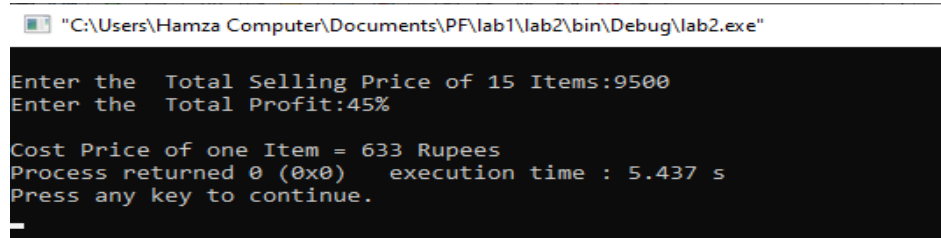
If the total selling price of 15 items and total profit earned on them is input through the keyboard, write a program to find the cost price of one item.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int Total_S_P, Total_P, Cost_P;
    printf("\nEnter the Total Selling Price of 15 Items:");
    scanf("%d",&Total_S_P);

    printf("Enter the Total Profit:");
    scanf("%d", &Total_P);
    /* Find the Cost Price of One Item */
    Cost_P = Total_S_P - Total_P;
    Cost_P = Total_S_P / 15;
    printf(" \nCost Price of one Item = %d Rupees", Cost_P);
    return 0;
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
Enter the Total Selling Price of 15 Items:9500
Enter the Total Profit:45%

Cost Price of one Item = 633 Rupees
Process returned 0 (0x0) execution time : 5.437 s
Press any key to continue.
_
```